[Your College Logo]

Industrial Internship Report on

AgriSense AI: Crop Production Prediction and Crop/Weed Detection

Prepared by

SoubhagyaSPatil

Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time. My project was (Tell about ur Project) This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

TABLE OF CONTENTS

1 Preface....3
2 Introduction....4
2.1 About UniConverge Technologies Pvt Ltd....4
2.2 About upskill Campus....8
2.3 Objective....9
2.4 Reference....9
2.5 Glossary....10
3 Problem Statement....11
4 Existing and Proposed solution....12
5 Proposed Design/ Model....13
5.1 High Level Diagram (if applicable)....13
5.2 Low Level Diagram (if applicable)....13
5.3 Interfaces (if applicable)....13
6 Performance Test....14
6.1 Test Plan/ Test Cases....14
6.2 Test Procedure....14
6.3 Performance Outcome....14
7 My learnings....15
8 Future work scope....16

1 Preface

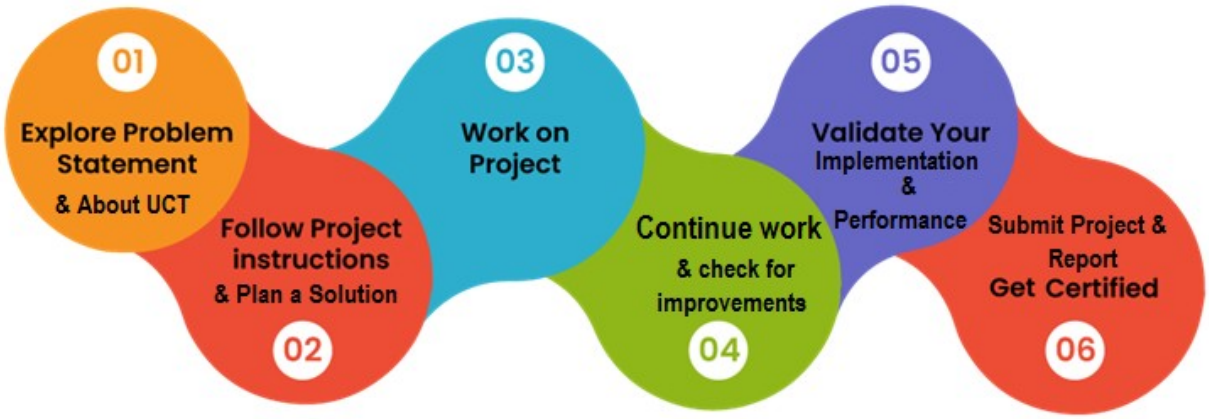
This report summarizes the AgriSense AI project work, which integrates historical crop production prediction and image-based crop/weed detection in a Flask web application.

The internship provides practical exposure to software development, data preprocessing, machine learning, computer vision, web application integration, debugging, documentation, and version control.

The project addresses two agricultural tasks: estimating crop production from historical data and identifying crop and weed objects in field images using a trained YOLO detector.

The internship report follows the supplied upskill Campus / The IoT Academy / UCT template and documents the project design, implementation approach, testing plan, learnings, and future scope.

The work is organized into requirement understanding, data preparation, model development, web integration, testing, improvement, and documentation.



The project helped develop skills in Python, Flask, scikit-learn, YOLO, SQLite, data handling, debugging, and technical report preparation.

I thank the internship organizers, faculty, mentors, and peers who supported the learning process. Add the names of your actual mentor and faculty guide before final submission.

My message to juniors is to build the project step by step, keep backups, test each module independently, document actual results, and verify all repository links before submission.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

UniConverge Technologies Pvt Ltd (UCT) is described in the supplied internship template as a company working in digital transformation and industrial solutions, using technologies such as IoT, cybersecurity, cloud computing, machine learning, communication technologies, Java, Python, and frontend development.

The template also introduces UCT solutions and platforms. This section is retained as background about the internship partner from the supplied template.

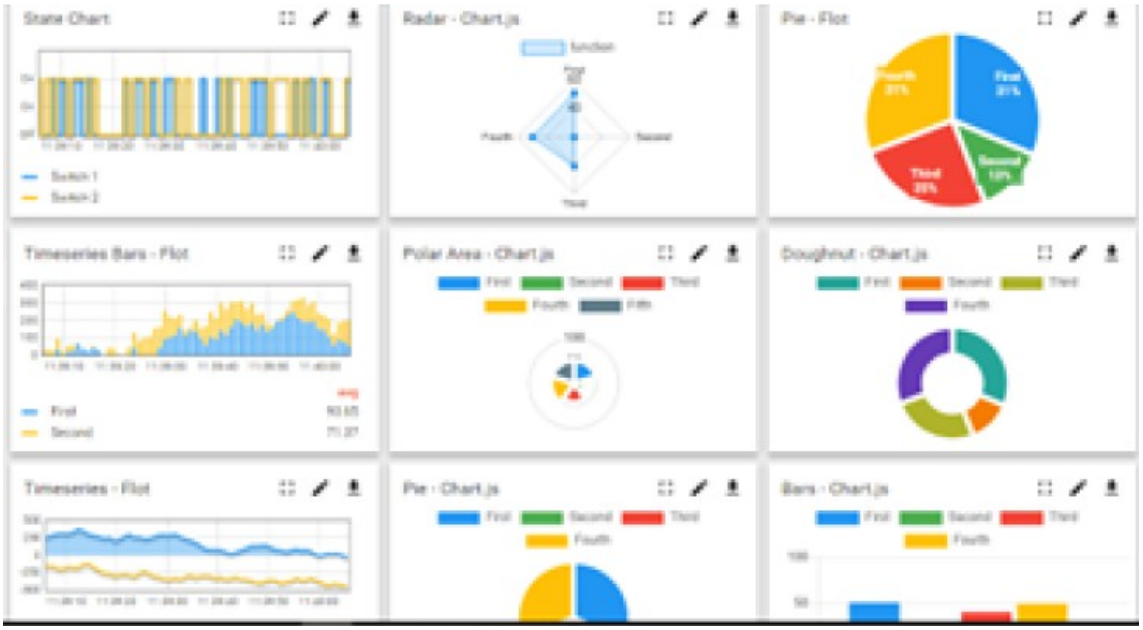


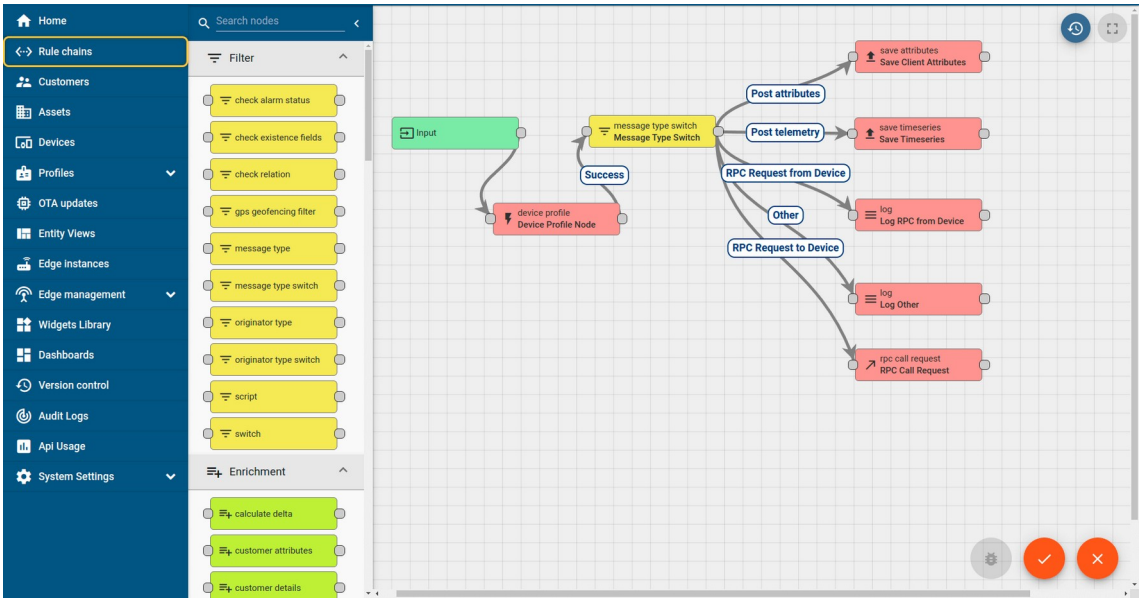
i. UCT IoT Platform (UCT Insight)

The template describes UCT Insight as an IoT platform designed for rapid deployment of IoT applications, with a Java backend, ReactJS frontend, and support for MySQL and NoSQL databases.

- The template lists connectivity through protocols such as MQTT, CoAP, HTTP, Modbus TCP, and OPC UA.
- The template states that UCT Insight supports cloud and on-premises deployment.

The template lists dashboard creation, analytics and reporting, alerts and notifications, third-party integration, and a rule engine as platform features.





ii. Smart Factory Platform (FACTORY WATCH)

The template describes Factory Watch as a platform for smart-factory requirements.

It supports production and asset monitoring.

- It can provide OEE and predictive maintenance capabilities.
- It helps users analyze machine-generated data and identify key performance indicators.
- Its modular architecture allows users to start with selected services and scale to more complex solutions.
- The template notes that the SaaS model is intended to save time and cost.

These descriptions are based on the supplied internship template and are included as partner background.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

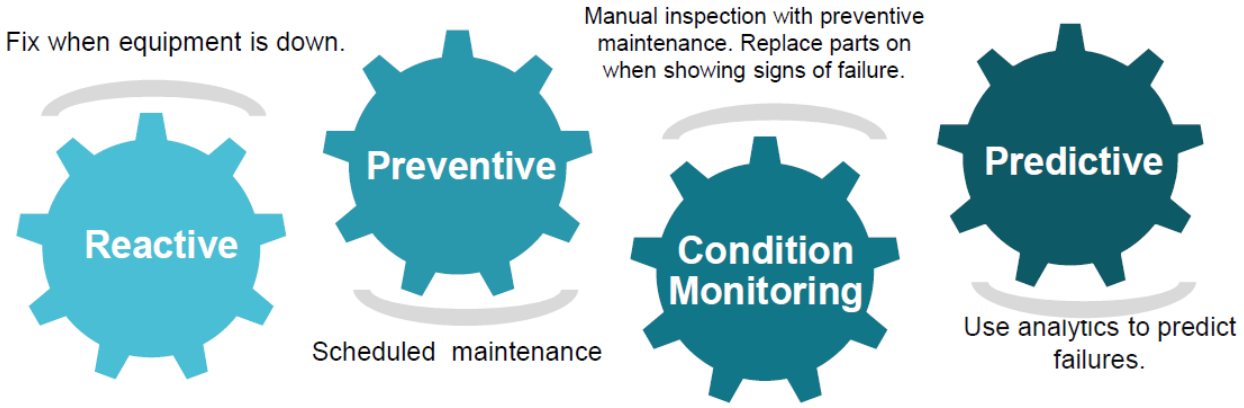


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

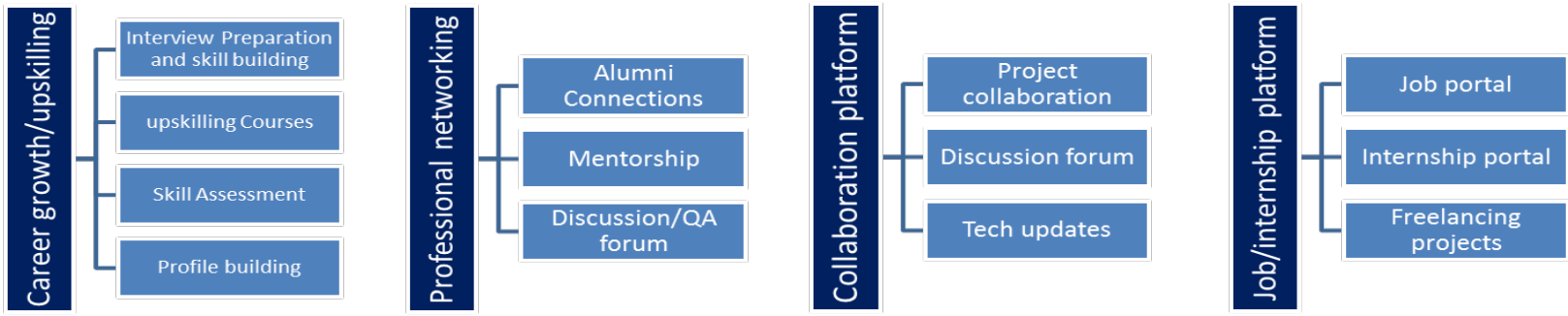
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.


Career growth/upskilling
Interview Preparation and skill building
upskilling Courses
Skill Assessment
Profile building
Professional networking
Alumni Connections
Mentorship
Discussion/QA forum
Collaboration platform
Project collaboration
Discussion forum
Tech updates
Job/internship platform
Job portal
Internship portal
Freelancing projects

2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- ☛ get practical experience of working in the industry.
- ☛ to solve real world problems.
- ☛ to have improved job prospects.
- ☛ to have Improved understanding of our field and its applications.
- ☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1]

[2]

[3]

2.6 Glossary

Terms	Acronym

3 Problem Statement

The project problem is to integrate historical crop production estimation and image-based crop/weed detection in one web application, while providing understandable results and maintaining a history of operations.

Agricultural production records and field-image inspection are different tasks. The application aims to accept a crop category and year for a baseline production estimate, and an uploaded image for object detection. Model reliability depends on appropriate datasets, correct labels, trained artifacts, and validation.

4 Existing and Proposed solution

Existing approaches include statistical analysis of historical crop data and computer-vision systems trained to detect objects in field images. Their limitations may include dependence on data quality, restricted coverage of crop types or regions, the need for labeled images, and difficulty accessing both functions through one interface.

AgriSense AI proposes a Flask application with two pipelines: a scikit-learn regression model for production estimates and a YOLO detector for crop/weed image analysis. SQLite is used to retain prediction and detection history.

The planned value addition is a unified dashboard, annotated detection images, saved history, clear model-readiness checks, consistent model paths, and future evaluation using suitable held-out data.

4.1 Code submission (Github link)

4.2 Report submission (Github link) : first make placeholder, copy the link.

4.3

5 Proposed Design/ Model

The system has two paths. Production: historical CSV → preprocessing → category/year features → regression model → Flask prediction route → SQLite history. Detection: labeled images → data.yaml → YOLO training → best weights → image upload → detected classes and bounding boxes → SQLite history.

5.1 High Level Diagram (if applicable)

Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)

5.3 Interfaces (if applicable)

Interfaces include the Flask dashboard, production prediction form, image upload form, annotated detection output, and history page. The model/data interfaces are a cleaned CSV for regression and YOLO image-label pairs plus data.yaml for detection.

6 Performance Test

Performance testing should assess correctness, model quality, response time, data validation, and reliability. Only measured results from actual runs should be reported; unexecuted tests are described as planned tests.

Key constraints include dataset completeness, correct class IDs, model-file availability, CPU/GPU training time, image size, upload format, and prediction response time.

The code uses separate training scripts, expected artifact paths, upload extension validation, a local SQLite database, and dashboard/history routes. The model path mismatch identified in the supplied scripts must be resolved before end-to-end detection testing.

No verified end-to-end test output or detector evaluation metrics were supplied with the code files reviewed for this report. Fill this section with screenshots and measured results after running the full application.

Relevant constraints include memory use, training time, inference speed, accuracy, image quality, data coverage, and availability of the trained model. If a constraint was not measured, state that clearly and explain its possible impact.

If testing is incomplete, report the test as pending rather than inventing results. Recommended next steps are to verify datasets and class IDs, train the models, test routes, record evaluation metrics, and inspect annotated images.

6.1 Test Plan/ Test Cases

6.2 Test Procedure

6.3 Performance Outcome

7 My learnings

The project strengthens practical skills in Python, data preprocessing, machine learning, computer vision, Flask application development, database integration, debugging, GitHub, and technical communication. These skills support future work in software engineering and applied AI.

8 Future work scope

Future work can include testing additional regression algorithms, adding reliable regional and weather features, evaluating YOLO with precision/recall and mAP metrics, improving authentication and deployment security, connecting live IoT sensor data, and collecting more diverse labeled images.